

Data, Signals, and Events Used in Platoon Formation Communication

The following table presents the data, signals, and events required for interaction and communication between trucks during platoon formation. It identifies the information exchanged, the communication messages, and the events that trigger system actions.

Type	Name	Description
Data	Truck ID	Unique identifier of joining truck
Data	Position	Current vehicle position
Data	Speed	Current vehicle speed
Data	Target Gap	Desired inter-vehicle distance
Data	Platoon ID	Identifier of the platoon
Signal	Platoon Advertisement	Broadcast announcing platoon availability
Signal	Join Request	Request to join platoon
Signal	Approval Message	Authorization to join
Signal	Position Adjusted	Confirmation of correct position
Signal	Formation Complete	Notification that formation is finished
Event	Advertisement Received	Follower receives advertisement
Event	Join Request Sent	Follower initiates joining
Event	Approval Granted	Platoon manager approves request
Event	Position Reached	Desired gap achieved
Event	Platoon Formed	Formation completed

Communication Protocol for Platoon Formation

The platoon formation scenario requires reliable and low-latency communication between trucks and the platoon manager. Based on the sequence diagram, the communication involves periodic platoon advertisements, join requests, approval messages, formation commands, and status updates exchanged within strict timing constraints.

A suitable protocol for this scenario is 5G Cellular Vehicle-to-Everything (5G C-V2X) communication. 5G C-V2X supports direct Vehicle-to-Vehicle (V2V) communication for exchanging platoon-related messages and Vehicle-to-Infrastructure (V2I) communication for interaction with the platoon manager. The protocol provides low latency, high reliability, and sufficient bandwidth to support real-time coordination during platoon formation.

Communication Message	Protocol Type
Platoon Advertisement	5G C-V2X (V2V)
Join Request	5G C-V2X (V2V/V2I)
Join Approval	5G C-V2X (V2I/V2V)
Formation Commands	5G C-V2X (V2V)
Status Updates	5G C-V2X (V2V)
Road Status Information	5G C-V2X (V2I)